

# MATHerials

## Short guide to building the prototype

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August 26, 2026

### 1 Introduction

In this document we will shortly discuss how to replicate the prototype which is discussed in the report which can be found in the directory Documents in subdirectory Report. To build the prototype you will need the files in the directory named Hardware.

### 2 The box

In the subdirectory Laser cutting you can find two files. The file named Box.ai contains the outline of the box and the file Lid.ai contains the outline of the lid for this prototype. These files can be used to laser cut the box. For the first prototype 4mm triplex was used for both the box and the lid.

The lid can be attached to the box using hinges. The hinges need to be attached to the side of the little circular hole in the lid. The precise placement of these hinges on that particular side of the lid is left to the builder.

### 3 The feedback mechanism

#### 3.1 3D printed parts

In this subsection the names of the parts are the names of the step files.

Most of the parts for the feedback mechanism and the grips were 3D printed for the first prototype. The step files for the parts can be found in subdirectory 3D printing. For these parts I used eSun petg filament and a nozzle of 0.3 or 0.4 mm. For smaller parts such as the axle or the magnet holder I used a smaller nozzle for better results.

First you need to know how to assemble the mechanism underneath the circle in the lid. This mechanism includes the gearbox, axle, ballpulley left, handle, and the wheel. The axle goes through the holes in the gearbox. Underneath the gearbox the ballpulley (left) can be placed and the axle should go through the hole in the middle of the ballpulley. The wheel can be placed on top of the gearbox such that the axle goes through the middle of the wheel. The handle can be placed in the hole on the edge of the wheel.

The gearbox needs to be attached to the box. This can be done by drilling two holes in the box for 3.5 mm screws with a length of approximately 8 mm. Use the 3D printed gearbox, wheel, ballpulley wheel (left), and lid to determine where the gearbox needs to be attached and where to drill the holes in the box. The gearbox needs to be placed such that the handle in the wheel moves easily through the circle in the lid.

Once the gearbox is attached to the box, you can determine where placeholder for the ballpulley wheel (right) needs to be attached. The placeholder needs to be placed such that the ballpulley (right) is on the other end of the box compared to the gearbox and the ballchain needs to reach to the end of the line in the lid. In the prototype this placeholder is glued to the box with gorilla glue.

For the pulley a ballchain with a diameter of 2.4 mm is used. The ballpulley wheels are made to accommodate a ballchain with a 2.4 mm diameter. The ballchain needs to be spanned between the ballpulley wheels.

The handle in the line consists of three parts, namely linear handle lower and upper, and the step file named hall effect. This handle needs to be attached to the lid of the box. Linear handle upper goes on top of the box such that it fits into the line in the lid. The linear handle lower needs to be screwed onto the upper part and resides on the other side of the lid (the inside) with a screw of 3.5 mm and a length between 5 and 8 mm. Make sure that the hole in the linear handle lower faces away from the small hole in the lid. Now the hall effect (arm) needs to be screwed to the linear handle lower with a 3.5 mm screw with a length of around 5 mm. The hall effect (arm) needs to face away from the small hole in the lid.

Now the magnet holder needs to be placed on the ballchain. The magnet needs to be placed such that it is at the beginning of the line in lid when the handle in the wheel is just above the wooden line in the circle (in the lid). The magnet holder lower should be placed under the ballchain and the ballchain needs to go through it. Then the magnet holder upper needs to be placed on top of the lower part. The magnet holder upper has a circular hole in it. There a magnet should be placed.

### 3.2 The electrical circuit

For the electrical circuit a breadboard is used. This makes it easier to replace parts or change the circuit. The breadboard is glued to the lid next to the little hole in the lid.

The little hole in the lid is meant for an LED. The rectangular hole is for an on and off switch. A hall effect (sensor) needs to be glued to the arm extending from the handle attached to the lid. It needs to be glued such that the wires go in the direction of the LED. Moreover, you will need an energy source. In the first prototype four triple A batteries were used as the energy source for the electrical circuit. The electrical circuit should be assembled as represented in [Figure 1](#).

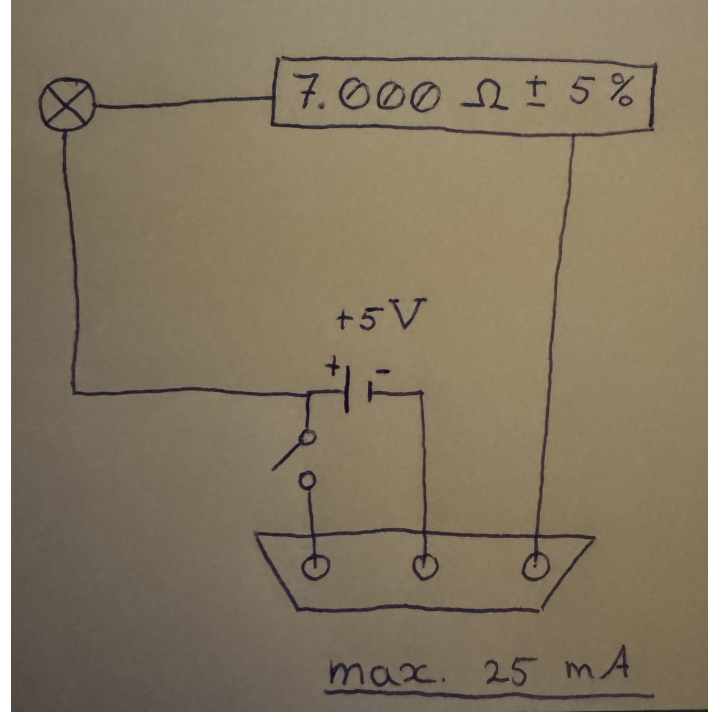


Figure 1: Representation of electrical circuit.